In [12]:
```
from sklearn import preprocessing
```

In [13]:
```
import pandas as pd
```

In [14]:
```
import numpy as np
```

In [15]:
```
df=pd.read_csv("IRIS.csv")
```

In [16]:
```
col_names=["sepal_length","sepal_width","petal_length","petal_width"]
```

In [17]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [19]:
```
min_max=preprocessing.MinMaxScaler()
```

In [27]:
```
df[col_names]=min_max.fit_transform(df[col_names])
df
```

Out[27]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	0.222222	0.625000	0.067797	0.041667	0
1	0.166667	0.416667	0.067797	0.041667	0
2	0.111111	0.500000	0.050847	0.041667	0
3	0.083333	0.458333	0.084746	0.041667	0
4	0.194444	0.666667	0.067797	0.041667	0
...	...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667	2
146	0.555556	0.208333	0.677966	0.750000	2
147	0.611111	0.416667	0.711864	0.791667	2
148	0.527778	0.583333	0.745763	0.916667	2
149	0.444444	0.416667	0.694915	0.708333	2

150 rows × 5 columns

In [21]:
```
df['species'].unique()
```

Out[21]:
```
array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

In [34]:
```
#lable_enc=preprocessing.LabelEncoder()
```

In [35]:
```
#df['species']=lable_enc.fit_transform(df['species'])
```

In [36]:
```
df
```

Out[36]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	0.222222	0.625000	0.067797	0.041667	0
1	0.166667	0.416667	0.067797	0.041667	0
2	0.111111	0.500000	0.050847	0.041667	0
3	0.083333	0.458333	0.084746	0.041667	0
4	0.194444	0.666667	0.067797	0.041667	0
...	...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667	2
146	0.555556	0.208333	0.677966	0.750000	2
147	0.611111	0.416667	0.711864	0.791667	2
148	0.527778	0.583333	0.745763	0.916667	2
149	0.444444	0.416667	0.694915	0.708333	2

150 rows × 5 columns

In [28]:

```
df
```

Out[28]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	0.222222	0.625000	0.067797	0.041667	0
1	0.166667	0.416667	0.067797	0.041667	0
2	0.111111	0.500000	0.050847	0.041667	0
3	0.083333	0.458333	0.084746	0.041667	0
4	0.194444	0.666667	0.067797	0.041667	0
...	...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667	2
146	0.555556	0.208333	0.677966	0.750000	2
147	0.611111	0.416667	0.711864	0.791667	2
148	0.527778	0.583333	0.745763	0.916667	2
149	0.444444	0.416667	0.694915	0.708333	2

150 rows × 5 columns

In [39]:

```
one_hot=preprocessing.OneHotEncoder()
```

In [40]:

```
encoded=one_hot.fit_transform(df[['species']]).toarray()
```

In [41]:
```
df
```

Out[41]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	0.222222	0.625000	0.067797	0.041667	0
1	0.166667	0.416667	0.067797	0.041667	0
2	0.111111	0.500000	0.050847	0.041667	0
3	0.083333	0.458333	0.084746	0.041667	0
4	0.194444	0.666667	0.067797	0.041667	0
...	...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667	2
146	0.555556	0.208333	0.677966	0.750000	2
147	0.611111	0.416667	0.711864	0.791667	2
148	0.527778	0.583333	0.745763	0.916667	2
149	0.444444	0.416667	0.694915	0.708333	2

150 rows × 5 columns

In [42]:
```
encoded_df=pd.DataFrame(encoded)
```

In [43]:
```
df
```

Out[43]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	0.222222	0.625000	0.067797	0.041667	0
1	0.166667	0.416667	0.067797	0.041667	0
2	0.111111	0.500000	0.050847	0.041667	0
3	0.083333	0.458333	0.084746	0.041667	0
4	0.194444	0.666667	0.067797	0.041667	0
...	...	...	...	...	...
145	0.666667	0.416667	0.711864	0.916667	2
146	0.555556	0.208333	0.677966	0.750000	2
147	0.611111	0.416667	0.711864	0.791667	2
148	0.527778	0.583333	0.745763	0.916667	2
149	0.444444	0.416667	0.694915	0.708333	2

150 rows × 5 columns

In [44]:
```
encoded_df
```

Out[44]:

	0	1	2
0	1.0	0.0	0.0
1	1.0	0.0	0.0
2	1.0	0.0	0.0
3	1.0	0.0	0.0
4	1.0	0.0	0.0
...	...	...	...
145	0.0	0.0	1.0
146	0.0	0.0	1.0
147	0.0	0.0	1.0
148	0.0	0.0	1.0
149	0.0	0.0	1.0

150 rows × 3 columns

In [45]:
```
encoded
```

```
Out[45]: array([[1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [1., 0., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
```

```
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 1., 0.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
```

```
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.],
       [0., 0., 1.]])
```

In [ ]:

In [2]:
```
import pandas as pd
import numpy as np
```

In [3]:
```
df=pd.read_csv("IRIS.csv")
```

In [4]:
```
df.shape
```

Out[4]:
```
(150, 5)
```

In [5]:
```
df.columns
```

Out[5]:
```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')
```

In [6]:
```
df.head()
```

Out[6]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [7]:
```
df.head(n=10)
```

Out[7]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
5	5.4	3.9	1.7	0.4	Iris-setosa
6	4.6	3.4	1.4	0.3	Iris-setosa
7	5.0	3.4	1.5	0.2	Iris-setosa
8	4.4	2.9	1.4	0.2	Iris-setosa
9	4.9	3.1	1.5	0.1	Iris-setosa

In [10]:
```
df.tail()
```

Out[10]:

	sepal_length	sepal_width	petal_length	petal_width	species
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

In [11]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [12]:

```
df.describe()
```

Out[12]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [14]:

```
df['species']
```

Out[14]:
```
0         Iris-setosa
1         Iris-setosa
2         Iris-setosa
3         Iris-setosa
4         Iris-setosa
              ...
145    Iris-virginica
146    Iris-virginica
147    Iris-virginica
148    Iris-virginica
149    Iris-virginica
Name: species, Length: 150, dtype: object
```

In [15]:
```
df.isnull()
```

Out[15]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	False	False	False	False

150 rows × 5 columns

In [16]:
```
df.isnull().sum()
```

Out[16]:
```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

In [17]:
```
df.isnull().sum().sum()
```

Out[17]:
```
np.int64(0)
```

In [18]:
```
df.dtypes
```

Out[18]:
```
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species          object
dtype: object
```

In [19]:
```
df.columns.values
```

Out[19]:
```
array(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'], dtype=object)
```

In [20]:
```
df.describe(include='all')
```

Out[20]:

	sepal_length	sepal_width	petal_length	petal_width	species
count	150.000000	150.000000	150.000000	150.000000	150
unique	NaN	NaN	NaN	NaN	3
top	NaN	NaN	NaN	NaN	Iris-setosa
freq	NaN	NaN	NaN	NaN	50
mean	5.843333	3.054000	3.758667	1.198667	NaN
std	0.828066	0.433594	1.764420	0.763161	NaN
min	4.300000	2.000000	1.000000	0.100000	NaN
25%	5.100000	2.800000	1.600000	0.300000	NaN
50%	5.800000	3.000000	4.350000	1.300000	NaN
75%	6.400000	3.300000	5.100000	1.800000	NaN
max	7.900000	4.400000	6.900000	2.500000	NaN

In [21]:
```
df.sort_index(axis=1,ascending=False)
```

Out[21]:

	species	sepal_width	sepal_length	petal_width	petal_length
0	Iris-setosa	3.5	5.1	0.2	1.4
1	Iris-setosa	3.0	4.9	0.2	1.4
2	Iris-setosa	3.2	4.7	0.2	1.3
3	Iris-setosa	3.1	4.6	0.2	1.5
4	Iris-setosa	3.6	5.0	0.2	1.4
...	...	...	...	...	...
145	Iris-virginica	3.0	6.7	2.3	5.2
146	Iris-virginica	2.5	6.3	1.9	5.0
147	Iris-virginica	3.0	6.5	2.0	5.2
148	Iris-virginica	3.4	6.2	2.3	5.4
149	Iris-virginica	3.0	5.9	1.8	5.1

150 rows × 5 columns

In [22]:
```
df.sort_values(by="sepal_length")
```

Out[22]:

	sepal_length	sepal_width	petal_length	petal_width	species
13	4.3	3.0	1.1	0.1	Iris-setosa
42	4.4	3.2	1.3	0.2	Iris-setosa
38	4.4	3.0	1.3	0.2	Iris-setosa
8	4.4	2.9	1.4	0.2	Iris-setosa
41	4.5	2.3	1.3	0.3	Iris-setosa
...	...	...	...	...	...
122	7.7	2.8	6.7	2.0	Iris-virginica
118	7.7	2.6	6.9	2.3	Iris-virginica
117	7.7	3.8	6.7	2.2	Iris-virginica
135	7.7	3.0	6.1	2.3	Iris-virginica
131	7.9	3.8	6.4	2.0	Iris-virginica

150 rows × 5 columns

In [23]:
```
df.iloc[5]
```

Out[23]:
```
sepal_length            5.4
sepal_width             3.9
petal_length            1.7
petal_width             0.4
species         Iris-setosa
Name: 5, dtype: object
```

In [24]:
```
df[0:3]
```

Out[24]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa

In [25]:
```
df.loc[: ,["sepal_length","sepal_width"]]
```

Out[25]:

	sepal_length	sepal_width
0	5.1	3.5
1	4.9	3.0
2	4.7	3.2
3	4.6	3.1
4	5.0	3.6
...	...	...
145	6.7	3.0
146	6.3	2.5
147	6.5	3.0
148	6.2	3.4
149	5.9	3.0

150 rows × 2 columns

In [26]:
```
df.iloc[:7 ,:]
```

Out[26]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
5	5.4	3.9	1.7	0.4	Iris-setosa
6	4.6	3.4	1.4	0.3	Iris-setosa

In [27]:
```
df.iloc[: ,:8]
```

Out[27]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [28]:

```
df.iloc[:1 ,:10]
```

Out[28]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa

In [29]:

```
df.iloc[3:5 ,0:2]
```

Out[29]:

	sepal_length	sepal_width
3	4.6	3.1
4	5.0	3.6

In [31]:

```
df.iloc[[1,2,4],[0,2]]
```

Out[31]:

	sepal_length	petal_length
1	4.9	1.4
2	4.7	1.3
4	5.0	1.4

In [32]:

```
df.iloc[1:3 ,:]
```

Out[32]:

	sepal_length	sepal_width	petal_length	petal_width	species
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa

In [ ]: